Epigenesis: changing DNA and simulating the consequences for neural development

Deyan Todorov

Technical paper, draft 3 · 13 November 2026 (Epigenesis 1.7.0) · Bulgarian version: epigenesis-technical-paper.bg.pdf
Code: `https://github.com/dttdrv/epigenesis` (Apache-2.0)

Abstract

Epigenesis is software for changing a piece of DNA and observing what follows in a simulated developmental system. It reads a DNA source under an explicit interpretation of what the DNA means, compiles it into a content-addressed Development Module, and replays every compilation with a separate validator. This paper describes the compiler, its verification architecture, and two coupled neural workflows introduced through Version 1.7.0. First, a mechanistic pipeline connects a DNA site—and in 1.7, arbitrary-length human enhancers through strand-symmetric Boltzmann partition physics across Hallikas GLI2/GLI3, 532 JASPAR 2024 vertebrate motifs, or custom matrices—to arbitrary N-gene regulatory differential equations compiled in sealed `brainc` v2/v3 modules. In a 9-base *Nkx2.2* GLI site, the substitution GACCACCCA to GAACACCCA shifts simulated Nkx2.2 from 0.940 to 0.0085 while passive promoter occupancy changes only from 0.940 to 0.928, and reverting the base restores the reference trajectory exactly. Scanning a full 171-base human enhancer (`chr15:90936956–90937127`) exposes 7 cryptic bifurcations among 513 substitutions (33 across five GSE188264 loci), epistatic second-site fate rescues on suboptimal enhancers, and 74 cryptic transitions in a multi-TF three-lineage circuit. Second, two CPU sequence predictors estimate variant effects on reporter assays: a Huber-linear model on SK-N-SH data (internal MSE 3.02% below 5-mers, yet external cross-study correlation 0.049 and MSE 9.52% above zero effect) and an orthogonal 0–72 h H1 neural-induction motif-temporal ridge predictor (`motif_temporal`, retrospective held-out family MSE 0.252079 against 0.285185 for 5-mers across 118 families). Epigenesis Studio (v0.2.0) and a unified CLI run these workflows locally in English and Bulgarian and export canonical SHA-256 receipts for independent replay.

Keywords: DNA compiler, regulatory variants, Boltzmann partition affinity, cryptic bifurcations, neural development, reproducibility, variant-effect prediction.

1 Introduction

A researcher who changes a regulatory DNA site wants to know what the change does to a developing system, ideally before choosing a laboratory experiment. Answering takes a chain of steps: which bases change, how the change alters binding, how that alteration enters a dynamical model, and how the model responds over time. In practice the chain lives in scripts that do not record their assumptions. A result then cannot be traced to the source bytes it came from, and another person cannot reproduce it.

Epigenesis treats a DNA record as the source of a program. A source profile fixes the grammar and identity of the record. The caller supplies an interpretation that states what the DNA is taken to mean. The compiler lowers both to a typed Development Module, an independent validator replays the derivation, and executable models read the sealed module. A compilation is accepted only if that second implementation reproduces it.

The paper follows the order in which a researcher meets the software. Section 3 describes the compiler, the validator, the unified CLI and the executable experiments. Section 4 presents the mechanistic chain: first the 9-base GLI worked example, then its Version 1.7 generalization to full-length enhancers, multi-motif Boltzmann binding, arbitrary N-gene circuits, epistatic second-site rescues and structural tissue bridging. Section 5 describes the sequence predictors—the Siraj Huber-linear checkpoint with its cross-study limits and the Kreimer 0–72 h neural-induction motif-temporal model. Section 6 describes Epigenesis Studio, which runs all workflows, and Section 7 states the limits.

Among the works reviewed in Section 2, none compiles existing DNA under an explicit interpretation with independent replay. This is a scoped comparison; we ran no head-to-head benchmark against other tools. The compiler does not derive biology from sequence, because the interpretation is an input. The simulations hold under their stated assumptions and do not predict embryo outcomes.

2 Related work

Compilers for DNA. Several systems compile a design description into DNA or genetic circuits: attribute grammars (Cai et al. 2009), a spatial language compiled to regulatory networks (Beal et al. 2011), the Eugene and Genotype Specification languages (Bilitchenko et al. 2011; Wilson et al. 2016) and Cello (Nielsen et al. 2016). Epigenesis runs the other way. It takes existing DNA and an explicit interpretation and lowers them to a model input.

Simulated development. CX3D grows neurons and networks from local rules (Zubler and Douglas 2009). An instruction language drives it (Zubler et al. 2011) and was applied to cortical development (Zubler et al. 2013). Two experiments here ('neural-construction' and 'neural-circuit' bridging) use CX3D.

Reproducible models. COMBINE archives package a model with its data and metadata (Bergmann et al. 2014). Epigenesis is narrower. It fixes what a derivation from DNA must record.

Biological models in the experiments. The Sonic Hedgehog readout in the neural tube is a well-quantified model of cell-fate decisions (Balaskas et al. 2012; Cohen et al. 2014; Herrera-Delgado et al. 2020). Enhancer evidence comes from Peterson et al. (2012) and binding affinities from Hallikas et al. (2006).

Variant-effect prediction. Section 5 and Appendix B use reporter-assay data (Melnikov et al. 2012; Tewhey et al. 2016; Kircher et al. 2019; Kreimer et al. 2022) and compare against sequence models (Zhou and Troyanskaya 2015; Kelley et al. 2016, 2018; Avsec et al. 2021; Chen et al. 2022). The delivered predictors combine regularized regression (Huber Huber 1964 or orthogonal temporal ridge) with JASPAR 2024 motif scores (Rauluseviciute et al. 2024). Both methods are established. What we contribute is the implemented, audited and replayable workflow around them.

3 The software

3.1 Compiler and contracts

The compiler turns a DNA source into a Development Module in four stages:

$$\text{DNA source} \rightarrow \text{typed source bundle} \rightarrow \text{explicit interpretation} \rightarrow \text{Development Module}$$

Built-in profiles read raw IUPAC sequence, FASTA, streaming reference FASTA, GenBank and GFF3. Other grammars enter through a version-2 external frontend. This frontend is an executable pinned by digest and paired with a separate executable validator. The caller supplies the interpretation (manifest, request, response), a lowering policy and a target contract. The compiler checks types, coordinates, references and budgets. Version 1.7 adds `brainc` v3 bundle contracts alongside v1/v2 for multi-transcription-factor circuit pairs and arbitrary-length motif-temporal predictions while preserving every v1/v2 schema and hash identity, and ships a standard-library unified CLI (`epigenesis`, `epigenesis_cli.py`).

Every artifact is a closed JSON object. At every level, the compiler rejects unknown or missing members, duplicate keys, non-finite numbers, integers beyond $\pm(2^{53}-1)$ and wrong types. Identities are SHA-256 digests of the RFC 8785 canonical encoding (Rundgren et al. 2020; NIST 2015). Sequence identities follow the GA4GH refget scheme. `SPEC.md` states the contracts. `docs/reference/STANDARDS.md` maps each contract to its specification.

Table 1. Executable experiments.

Experiment	What it does	Result and assumption
Neural patterning	Reads a 9-bp *Nkx2.2* enhancer substitution (Peterson et al. 2012), measured GLI binding ratios (Hallikas et al. 2006) and a four-gene model (Herrera-Delgado et al. 2020; Cohen et al. 2014)	Reversion restores the original state. Signal-history, rescue and phenocopy controls and numerical refinement pass. In-vitro binding ratio is transferred to intracellular affinity by assumption.
Neural circuit	Scans arbitrary enhancers via strand-symmetric Boltzmann physics (GLI2/GLI3, 532 JASPAR motifs, or custom PWMs) into N-gene ODEs (`neural-tube`, `mutual-toggle`, `tri-lineage-switch`, or custom `CircuitSpec`), tests second-site rescue and optionally bridges to CX3D	Exposes 7 cryptic bifurcations on `chr15` (33 across 5 loci) and 74 on `tri-lineage-switch`; exact 9-bp parity ($< 10^{-15}$) and reversion (0.0). Conditional on partition-sum affinity transfer and uncalibrated model units.
Neural regulation	H1 neural-induction 0–72 h reporter assay (Kreimer et al. 2022): frozen k-mer baseline (`regulation.py`) and motif-temporal model (`motif_temporal.py`)	k-mer baseline failed dinucleotide gate; `motif_temporal` reaches held-out family MSE 0.252079 vs 0.285185 (k5) across 118 families, conditional on published preprocessing.
Coding consequences	Reconstructs *Sox2* edits from genomic FASTA and translates them with the NCBI standard code	A four-base deletion gives a 97-residue product (69 unchanged residues) and a stop at codon 98, matching Gong et al. (2020). No protein abundance is claimed.
Neural evidence	Reconstructs 109 observations of *gcy-22* expression after CHE-1 depletion in *C. elegans* (Traets et al. 2021), traced to workbook cells and DNA edits	Mean auxin/control ratios are 8.70% for a transferred 12-base core and 66.33% for a larger fragment. Sampling limits and conflicting labels stay attached.
Neural fate	Computes Pax6, Olig2 and Nkx2.2 dynamics from a 512-base artificial parameter recipe	Reproduces specified behavior of published ODE models. The equations set the interactions. The recipe does not discover them.
Neural construction	Grows cells, neurites and contacts from one founder and an artificial 480-base recipe (15 parameters) on CX3D; stimulates cells and lesions contacts	Morphology and connections emerge during execution and are not encoded. The recipe is artificial.

3.2 Independent validation

A producer-isolated validator reparses the original source, reruns external validators on fresh snapshots and replays lowering and linking. It does not import the producer. The Version 1.7 release suite runs 535 unit and integration tests (four skip when optional Java/external tools are absent), 32 Studio tests, and end-to-end release gates (`tests/verify.py`) covering the minimal example, universal translation, boundary conditions, contract attacks, path safety, real-DNA compilation, `neural-circuit`, `neural-regulation-motif`, `neural-construction`, and wheel and source-archive installation. The minimal example reproduces four fixed digests: source descriptor, Development Module, development bundle and validation report.

3.3 Executable experiments

Each experiment states its inputs, assumptions, controls and the claim it supports (Table 1). Neural patterning and neural circuit form the mechanistic chain of Section 4.

Two earlier experiments ended in negative results and stay in the repository (Appendix B). The experiments show that the chain from DNA through an explicit interpretation to a mechanistic model can be

Table 2. Worked example at 20 model units: GLI activator fraction 0.2, GLI2 profile, all genes starting at zero. Values are concentrations in model units; occupancy is the probability that the *Nkx2.2* promoter is active.

Run	Pax6	Olig2	Nkx2.2	Irx3	Occupancy
Reference	0.00519	0.0208	0.940	0.000716	0.940
Variant, edit inside the network	0.241	0.967	0.00846	0.00206	0.00846
Variant, passive reporter	0.00519	0.0208	0.940	0.000716	0.928
Reversion of the base	identical to the reference; maximum absolute difference 0				

executed and audited. They do not show that the interpretations are right.

4 Worked example: changing a GLI site

4.1 The question and the chain

The question is how a specified DNA change alters the dynamics of a neural-development model. The example is the experiment `examples/neural-patterning`. Each step below is a separate piece of data or code in the repository, and each assumption is declared.

1. **Site.** Peterson et al. (2012) tested a substitution in a GLI binding site of the *Nkx2.2* -2 kb enhancer. In the strand orientation of the binding table the nine-base core changes from GACCACCCA to GAACACCCA, a C to A change at position 3. The repository holds only these cores; the full enhancer is not included.
2. **Binding.** Hallikas et al. (2006) measured GLI2 and GLI3 binding to every single-base substitution of the consensus. The mutant-to-reference ratio q of this substitution is 0.0424 for GLI2, the primary choice, and 0.0353 for GLI3, a sensitivity check. The compiler turns the selected site into a Development Module that stores only the selected nucleotides (the tensors `count` and `nucleotides`). The simulation reads those bases back from the validated module and derives q from the binding table.
3. **Transfer.** By declared assumption, q multiplies the GLI affinity of the *Nkx2.2* promoter in the four-gene model of Herrera-Delgado et al. (2020), whose competitive-binding mechanism follows Cohen et al. (2014). Let S be the fraction of GLI in its activator form, w the basal weight, K the GLI coefficient, c the activation coefficient and D the cross-repression factor. The promoter is active with probability P, its occupancy:

$$P = \frac{w(1+K)(1+qcKS)}{w(1+K)(1+qcKS) + (1+qK)D}, \qquad \frac{dX}{dt} = 2P - 2X,$$

   where X is the concentration of the gene product in model units. For the reference, $q = 1$ and the factor $1+K$ cancels, which recovers the published equations. At a fixed endogenous state, a weaker site lowers occupancy for signals above $S = 1/c = 0.1$ and raises it below.
4. **Dynamics.** The four genes Pax6, Olig2, Nkx2.2 and Irx3 are integrated by adaptive fourth-order Runge–Kutta steps under the signal history and starting state that the user supplies. Concentrations and time are in model units.

4.2 Reference, variant and reversion

We ran the reference and the variant under one protocol: a GLI activator fraction of 0.2 for 20 model units, with all four genes starting at zero. The variant runs in two ways. As an edit inside the network (endogenous mode), the changed affinity feeds back into all four genes. As a passive reporter (reporter mode), the promoter reads out occupancy on the unchanged reference network. A third run reverts the base and repeats the reference.

With the weaker site, Nkx2.2 inside the network ends at 0.0085 where the reference ends at 0.940. The other genes follow. Nkx2.2 represses Pax6, Olig2 and Irx3 in this model, so less Nkx2.2 releases them: Olig2 rises from 0.021 to 0.967 and Pax6 from 0.0052 to 0.241. Read through a passive reporter, the

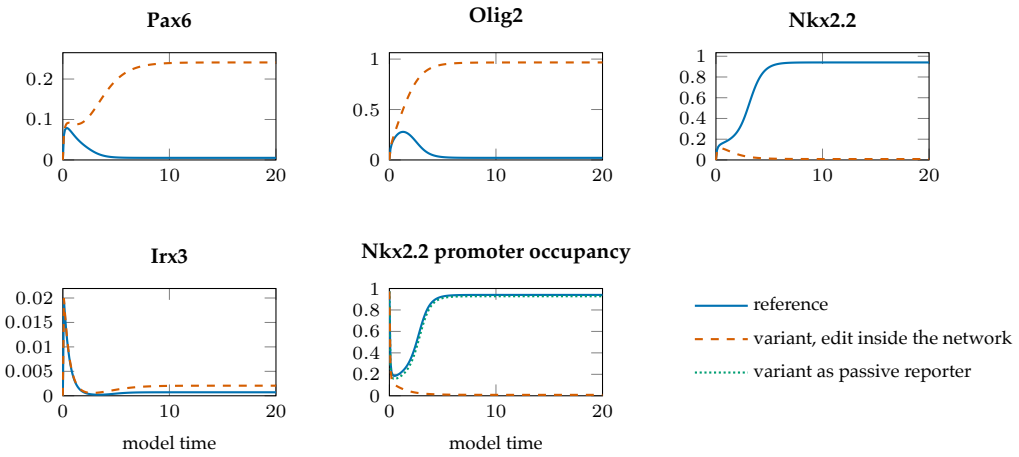


Figure 1. Trajectories behind Table 2. In the four gene panels, the dashed line is the variant as an edit inside the network. In the occupancy panel, the dotted line is the variant as a passive reporter, which stays close to the reference (0.928 against 0.940 at the end). Reporter mode leaves all four genes on the reference trajectory.

same substitution moves promoter occupancy only from 0.940 to 0.928 and leaves all four genes on the reference trajectory. A reporter assay would therefore show a small change for an edit whose effect inside the feedback network is large. With GLI3 in place of GLI2, the variant ends with Nkx2.2 at 0.0081.

Reverting the base gives a trajectory identical to the reference at all 201 output times and four genes (maximum absolute difference 0), and the reverted site compiles to the same Development Module as the reference. The backend adds twelve mechanism checks, and all pass in the demonstration run. A mediator rescue keeps the variant DNA and imposes the reference multiplier. A phenocopy keeps the reference DNA and imposes the variant multiplier. A signal-history control shows that identical final signals can leave different Nkx2.2 states. Further checks cover the sign reversal of the binding effect across signals and convergence when the output interval is halved (differences below 10^{-4}).

4.3 Assumptions and scope

The result depends on six declared assumptions, listed in `examples/neural-patterning/data.json` and shown with every Studio run.

1. The published normalized binding ratio is transferred as a relative affinity multiplier. The original normalization pipeline is not fully recovered.
2. The same GLI profile scales activator and repressor affinity. GLI2 is primary and GLI3 is a sensitivity analysis.
3. The nine-base site is assigned to a published *Nkx2.2* regulatory context. The rest of the enhancer, chromatin and cofactors are fixed and unmodeled.
4. Reporter mode reads promoter occupancy on an unchanged endogenous trajectory. Endogenous mode is a hypothetical genomic edit with feedback.
5. The signal is the GLI activator fraction in $[0, 1]$. It is not a calibrated Shh concentration.
6. Time and concentration are in model units. No fluorescence scale, detection threshold, biological clock or embryo probability is inferred.

Peterson et al. (2012) report *Nkx2.2* reporter expression in 4 of 6 embryos with the reference site and in 0 of 13 with the variant (E10.5). The model has no detection threshold or embryo-level noise model, so these counts are shown beside the simulation and are not scored as a reproduced prediction. The four-gene model omits Foxa2.

The supported 9-bp path takes one substitution with a measured, positive binding weight in this site, a signal protocol of up to 10 phases and 100 model units, and an optional starting state. Version 1.7 lifts this restriction via `examples/neural-circuit`.

Table 3. Summary of Version 1.7 full-enhancer and multi-circuit scans (`neural-circuit`, $S = 0.20$ for neural-tube and $S = 0.35$ for tri-lineage).

Enhancer / benchmark	Circuit & TF(s)	Length	Cryptic	Variants	Second-site rescue
`chr15:90936956-90937127`	`neural-tube` (GLI2)	171	7	513	None (single core)
Five GSE188264 loci (benchmark)	`neural-tube` (GLI2)	171	33	2,565	`chr18`, `chr1`
`chr15:90936956-90937127`	`tri-lineage-switch` (`MA0069.1`+`MA0463.3`)	171	74	513	Supported in API

4.4 Generalizing to full enhancers and arbitrary N-gene circuits

To move beyond a single 9-base consensus site and a single fixed four-gene diagram, `neural-circuit` links arbitrary regulatory sequences directly to user-defined N-gene differential equation circuits (`CircuitSpec`, 1 to 64 genes). For any transcription-factor binding matrix M—either the Hallikas GLI2/GLI3 profiles, any of the 532 JASPAR 2024 vertebrate matrices, or a custom position-weight matrix—the partition sum over all forward and reverse-complement sliding windows is

$$Z(S_{\mathrm{DNA}}, M) = \sum_{i=1}^{L-W+1} \frac{w_{+,i} + w_{-,i}}{2} + \varepsilon, \qquad q = \frac{Z(S_{\mathrm{var}}, M)}{Z(S_{\mathrm{ref}}, M)},$$

with default Laplace pseudocount $\varepsilon = 10^{-4}$ (at $\varepsilon = 0$ on the 9-bp site, GAACACCCA reproduces the measured 0.0423555 ratio to within 10^{-15}). Ratios bind per gene as activators (q) or repressors ($1/q$) under either competitive or activation-only occupancy, and pairs compile with exact sequence reversions into sealed `brainc` v2 or v3 Development Modules supporting SNVs, MNVs and indels.

Across real human 171-base enhancers from GSE188264 (Table 3), scanning all $3L = 513$ single-base substitutions exposes *cryptic bifurcations*: variants that alter passive promoter occupancy by less than 0.02 yet cross a saddle-node threshold in endogenous feedback to flip final cell fate. On `chr15:90936956-90937127` under GLI2 ($S = 0.20$), 7 substitutions are cryptic bifurcations (including `60:C>A`, $q = 0.042455$, passive reporter -0.0124, endogenous Nkx2.2 -0.9318, $75.2\times$ amplification and spatial P3 domain contraction from $x = 0.25$ to 0.125). Across five curated GSE188264 enhancers, 33 substitutions act as cryptic bifurcations, and suboptimal enhancers (`chr18:56475717-56475888` and `chr1:175099416-175099587`) admit second-site epistatic substitutions that restore the wild-type fate without reverting the primary mutation. Applying the same framework to a multi-factor three-lineage circuit (`tri-lineage-switch`: `MA0069.1` PAX6 activator on `Excitatory` paired with `MA0463.3` BCL6 repressor on `Inhibitory`) yields 74 cryptic fate transitions on `chr15` (e.g. `24:T>C`, shifting `Excitatory` to `Progenitor`, $\Delta = 0.8149$ vs. passive 0.0). Steady states can optionally lower into the 480-base CX3D construction recipe for morphogenetic and LIF transmission checks. Every generalized circuit result remains conditional on affinity-transfer assumptions and uncalibrated units (`biological_acceptance = False`).

5 The edit predictor

The simulation of Section 4 follows one chosen change through a mechanistic model. The predictor looks at many candidate edits at once. For each, it estimates from sequence alone how much the edit changes reporter activity and in which direction. A researcher can use the scores to rank edits and to choose a panel to measure. Studio runs the predictor in its Scan and Edits tabs (Section 5.4).

5.1 Task and data

Task. Given a reference sequence and an edited sequence, estimate the alternate-minus-reference effect on reporter activity, a $\log_2$ ratio, in the SK-N-SH 24-hour assay of Siraj et al. (2026). Substitutions,

insertions and deletions are allowed. The score ranks edits within this assay. It is not a probability and it does not simulate development.

Screen and partitions. The Siraj screen has 365,762 paired designs, 336,475 of them with two 200-base alleles, measured with at least five replicate transfections per library. Edits are grouped into families, the connected components of overlapping or identical constructs (reverse complements included), so related edits never straddle a split. The delivered model is fitted on the FIT partition, 66,604 edits in 6,794 families, and compared on the internal TUNING partition, 16,986 edits in 1,759 families. A further RESERVE partition has not been scored and stays sealed. Development reused TUNING repeatedly, so results on it are tuning results (Appendix A lists the four classes of data).

External studies. Five neural reporter studies test transfer (Table 7). Each held-out set contains only variants with no variant of another study and no Siraj design within 2,000 bases, and it is the same for every model. Earlier models of this project had already seen these studies, so they are exposed data. A sixth study, Rummel et al., is a sensitivity analysis, because 83% of its variants have one allele with fewer than ten RNA reads (Appendix B).

5.2 Method

Features. For an edit, the model forms the difference between the edited and the reference sequence in two feature blocks (Figure 2). The first block counts the 512 canonical 5-mers, a 5-mer and its reverse complement counted together. The second has 532 motif features. For each JASPAR 2024 vertebrate motif (Rauluseviciute et al. 2024), the potential of a sequence is the sum over its windows of $\log(1+s\,r)$, where r is the odds of the motif at the window, averaged over both strands, and $s = 0.003$ compresses strong sites. The potential is measured against a window with $r = 1$. The feature is the potential of the edited sequence minus that of the reference. Only windows that overlap the edit differ, so the score depends on the 18 bases on either side of the change and on nothing farther away.

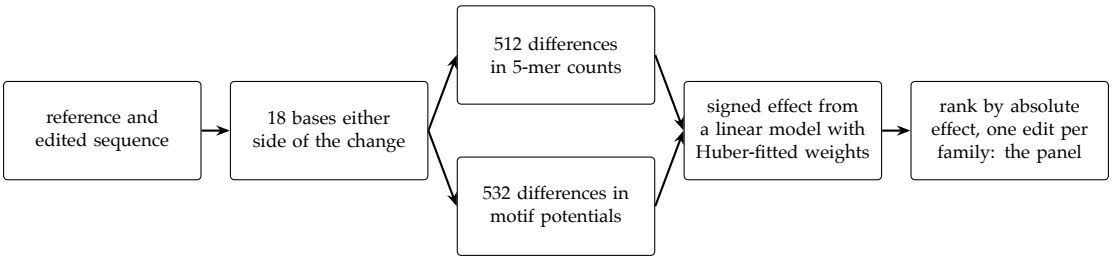


Figure 2. The predictor. An edit changes two feature blocks. A linear model turns them into a signed effect, and a panel takes the strongest edits, one per family. Studio runs this checkpoint.

Fit. Both blocks are fitted jointly, without an intercept, by minimizing a Huber loss (Huber 1964) with a penalty of 0.01 on the 5-mer block and 3 on the motif block, and with equal weight for every family. The Huber threshold is set once, from the median absolute deviation of the residuals of the squared-loss fit, and is not updated during optimization. Each fit converged in 10 to 13 updates. The checkpoint is `epigenesis-siraj-huber-linear-v1`.

Baselines. *k5* is ridge regression on the 5-mer block alone. *Joint-linear* fits both blocks with the same penalties and squared loss, so comparing it with Huber-linear isolates the loss. *No effect* predicts that every edit changes nothing. *S2*, used in Section 5.5, is the earlier Siraj-trained model that adds motif potentials to k5 (Appendix B).

Panels and utility. A panel of n edits is built by sorting the edits by absolute predicted effect (ties by identifier) and taking the first edit of each family not yet represented until n are chosen. The signed utility of a panel is the mean observed effect of its edits, each multiplied by the sign of its predicted effect. Random selection scores near zero. Intervals come from 2,000 resamples of whole families, and a lower bound is the 5th percentile of the gain over a comparator.

5.3 Results within the assay

Table 4 compares the model with its baselines on the internal tuning partition. Huber-linear has the lowest error, 3.02% below k5 and 5.40% below predicting no effect. Joint-linear differs from it only in the loss, and Huber-linear improves on it by 0.0030 with a positive lower bound (+0.0002). At panel sizes 50 and 200 Huber-linear has the highest utility (Figure 3). At ten edits joint-linear scores higher (1.601 against 1.483), because the two panels differ in two edits, and the data do not rank the models there.

Table 4. Internal tuning partition, 16,986 edits in 1,759 families. Error is the equal-family mean squared error (lower is better). Utility is the signed utility of a panel of 10, 50 or 200 edits (higher is better). The last row is the lower end of the 5th to 95th percentile interval of the gain of Huber-linear over k5 in 2,000 whole-family resamples.

Model	Error	Utility, 10	Utility, 50	Utility, 200
No effect	0.304075	0	0	0
k5	0.296611	0.857	0.607	0.509
Joint-linear (squared loss)	0.290683	1.601	0.954	0.626
Huber-linear (delivered)	0.287667	1.483	0.986	0.634
Gain over k5, lower bound	+0.0051	−0.150	+0.140	+0.063

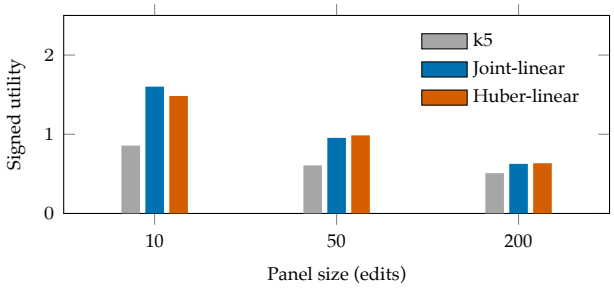


Figure 3. Signed utility of panels by size on the internal tuning partition (values in Table 4).

Six of the seven original requirements pass. The open one is the ten-edit gain over k5. Its point estimate is positive (+0.626), and its lower bound is negative (−0.150), so the data do not settle it. Development used this partition throughout, so we call the improvement preliminary.

5.4 Demonstration in Studio

Studio runs the delivered checkpoint. This demonstration uses two inputs that ship with it: a 200-base sequence, and a table of 64 edits in 11 families that carries no measured values, so every number below is a prediction. The script `docs/paper/figure-data.py` reproduces each value.

Scan. Pasting the 200-base sequence into the Scan tab scores all 600 single-base changes (Figure 4). Of the 600 edits, 343 are predicted to raise reporter activity and 257 to lower it, and the median absolute effect is 0.049. The largest increase is A to G at position 52 (+0.581) and the largest decrease is G to T at position 165 (−0.313). The first edit of the example table, G to A at position 100, scores −0.214 in Scan. The Edits tab returns the identical value when given the same edit as a pair, because a score depends only on the bases around the change.

Edits. The Edits tab takes pairs of a reference and an edited sequence, each with an optional family, and a panel size. For a panel of five on the 64-edit table, the five largest absolute scores include two edits of one family (the locus at chr10:102622941, +0.169 and +0.167). The selection rule keeps the first and passes to the next unrepresented family, so the panel has five edits from five families (Table 5). Each score is the sum of a 5-mer part and a motif part, which the receipt records separately.

Receipt. Exporting the Edits result writes a receipt. Loading it in the Verify tab, or running `epigenesis-`

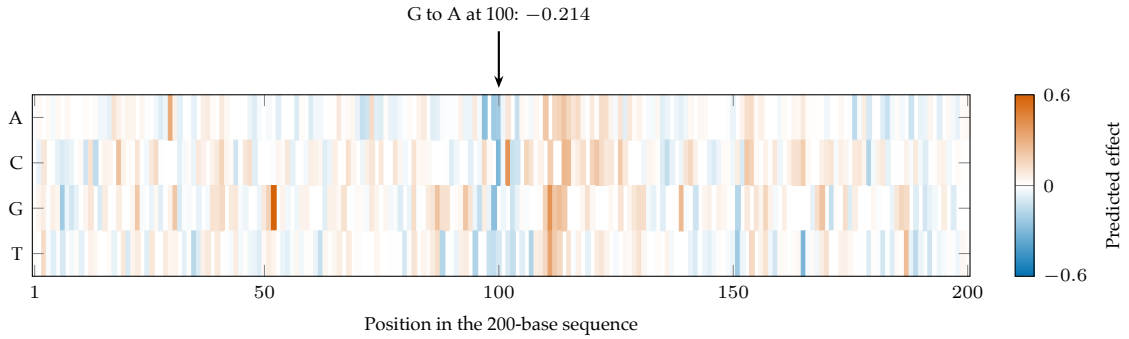


Figure 4. Studio Scan of the example sequence. Each column is a position and each row the base it changes to; the blank cell in a column is the reference base. Orange cells are predicted increases and blue cells decreases of reporter activity, saturating at ±0.6. The arrow marks the first edit of the example table.

Table 5. Panel of five from the 64-edit example table. Family is the label that the table gives each edit. Effect is the predicted alternate-minus-reference effect, the sum of the 5-mer part and the motif part.

Rank	Edit	Family	Effect	5-mer part	Motif part	Direction
1	chr10:101119242 G>A	101119242	−0.2140	−0.1261	−0.0878	lower
2	chr10:119062236 C>T	118615567	−0.2033	−0.1434	−0.0599	lower
3	chr10:102798346 T>C	102622941	+0.1687	+0.1407	+0.0280	raise
4	chr10:105307889 T>C	105307889	−0.1572	−0.1439	−0.0133	lower
5	chr10:119050998 C>T	119021407	+0.1326	+0.0162	+0.1164	raise

`studio verify`, replays the request and passes all three checks, with a maximum difference of 0 between stored and replayed numbers. Table 6 shows altered copies: a changed result fails the replay even when the digest is recomputed, and a renamed model fails the identity check. A receipt tells a reader that a result came from this checkpoint and this request. It says nothing about whether the prediction is right.

Table 6. Verify on the receipt of the panel above and on three altered copies. The last column gives the largest difference between stored and replayed numbers.

Receipt	Digest	Model identity	Replay
Exported	pass	pass	pass (0)
First stored effect raised by 0.001	fail	pass	fail (0.001)
The same, digest recomputed	pass	pass	fail (0.001)
Model renamed, digest recomputed	pass	fail	pass (0)

5.5 Transfer to other studies

The same checkpoint was scored once, without refitting, on the five neural reporter studies (Table 7). Per-study correlations are small and differ in sign. The mean is 0.0490, against 0.0465 for S2 and 0.0460 for k5. The paired gain over S2 is +0.0024 with an interval of −0.0060 to +0.0109, so the correlations do not separate the models.

The error is the stricter test. Huber-linear has a lower mean squared error than every fitted comparator. Its mean of 0.1044 is nevertheless 9.52% above the 0.0953 of a predictor that says no edit has any effect, and each of the five per-study points and intervals favors no effect (Figure 5). Inside the assay the same model is 5.40% below that predictor. Study outcomes differ in definition and scale, and the no-effect comparison was declared after scoring, so this is a diagnostic and not a replication.

The scores rank edits slightly better than chance in these studies, and their scale does not carry from one assay to another. We make no claim of transfer to other assays, cell types or endogenous expression.

Thirteen frozen tests (Appendix C) found no model that reliably beat simple baselines across contexts.

Table 7. Huber-linear on the held-out edits of the five neural reporter studies, scored once without refitting. r is the family-weighted Pearson correlation. Error is the mean squared error of Huber-linear and of predicting no effect, and Excess is the first as a percentage above the second. Mean rows weight studies equally; the excess of the mean row compares the two mean errors. Matoba has 97 families. †Sensitivity study, outside the mean.

			Correlation r			Error		
Study	Edits	Families	Huber	S2	k5	No effect	Huber	Excess
Lee	5,916	2,340	0.0152	0.0185	0.0254	0.0626	0.0739	+18.10%
McAfee	1,682	676	0.0935	0.0910	0.0990	0.0277	0.0333	+20.38%
Matoba	4,013	97	0.0286	0.0225	0.0384	0.0164	0.0256	+56.53%
Guo	748	571	−0.0157	−0.0143	−0.0241	0.3479	0.3612	+3.83%
Weiss	9,247	6,930	0.1233	0.1149	0.0912	0.0220	0.0278	+26.46%
Mean of five			0.0490	0.0465	0.0460	0.0953	0.1044	+9.52%
Rummel†	611	349	0.0375	0.0488	0.0505	0.5420	0.5503	+1.53%

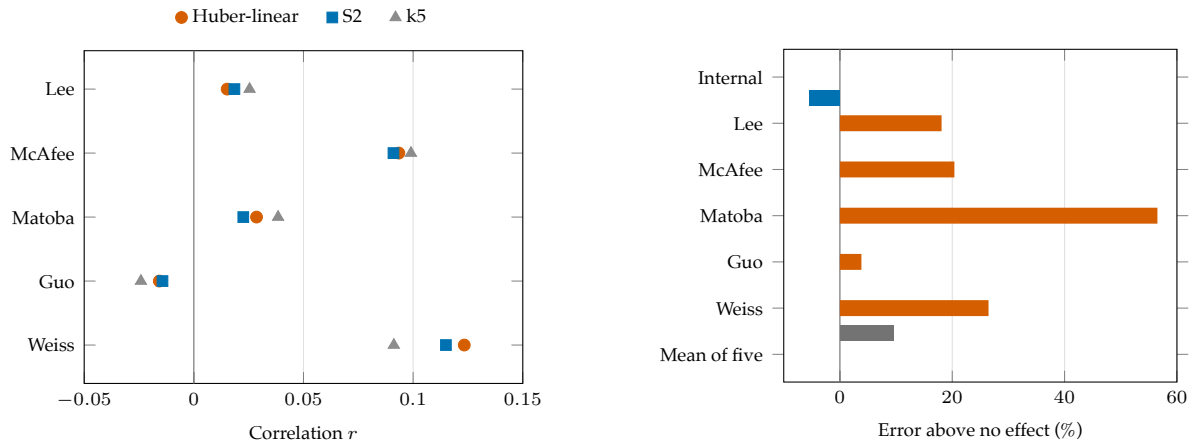


Figure 5. Left: correlation per study (Table 7). Right: error of Huber-linear relative to predicting no effect. The first bar is the internal tuning partition (below no effect), the next five bars are the studies, and the last bar compares the two mean errors.

5.6 Kreimer 0–72 h motif-temporal neural-induction predictor

A second Version 1.7 checkpoint, the Kreimer motif-temporal model (`epigenesis-kreimer-motif-temporal-v1`, implemented in `motif_temporal.py` inside `examples/neural-regulation/`), models temporal enhancer perturbations across seven timepoints ($[0, 3, 6, 12, 24, 48, 72]$ h) in the H1 neural-induction dataset of Kreimer et al. (2022) (GSE188264). Earlier frozen k-mer models (`regulation.py`) reduced squared error by 20.97% relative to zero effect yet failed to outperform a dinucleotide control (Appendix B). `motif_temporal` expands features to the same 512 canonical 5-mer differences plus 532 JASPAR 2024 motif potential differences ($s = 1.0$), decomposes the seven-hour response into an orthogonal time-averaged mean column and six temporal-deviation columns via QR factorization, and applies a $10\times$ stronger ridge penalty to the temporal-deviation columns (`word_penalty` $= 0.3$, `motif_penalty` $= 3.0$, `temporal_multiplier` $= 10.0$). All hyperparameters were selected purely on the training-to-validation family split before refitting on training plus validation.

Across 1,415 targeted perturbations in 118 held-out test families, equal-family test MSE drops to **0.252079** versus 0.285185 for the k5 sequence baseline, 0.319358 for dinucleotides, and 0.360870 for predicting no effect. The 95% whole-family bootstrap improvement intervals stay positive against every comparator: $[0.041258, 0.188584]$ over zero effect, $[0.008174, 0.138345]$ over dinucleotides, and $[0.007810, 0.061363]$ over k5. On time-centered trajectories, temporal-contrast MSE improves from 0.032108 to **0.031285** ($[0.000220, 0.001528]$). Arbitrary-length sequence pairs (5 to 16,384 bp, includ-

Table 8. Studio tabs in Version 0.2.0.

Tab	Input	Output
Simulate	One substitution of the nine-base site, GLI2 or GLI3, a signal protocol of up to 10 phases and 100 model units, and an optional starting state	Trajectories for reference, variant and reversion; reporter-versus-network comparison; controls; the six assumptions; CSV; receipt
Enhancer Circuit	Curated GSE188264 locus or FASTA enhancer (9 to 500 bases), circuit preset (`neural-tube`, `mutual-toggle`, `tri-lineage-switch`), GLI2/GLI3 or any of 532 JASPAR motifs, target gene(s), optional custom SNV/MNV/indel variant	Cryptic-bifurcation scan ($3L$ variants), endogenous vs. passive-reporter trajectories, 21-point spatial domain, pulse-chase derepression, second-site rescues, `brainc` v2/v3 receipts, Kreimer/Siraj side-scores
Scan	One FASTA record or plain sequence of 5 to 1,000 bases (A, C, G, T only; several records refused); Siraj or Kreimer 0–72 h toggle	Heatmap of all three substitutions at every position; 0–72 h profile and top JASPAR motifs (Kreimer); ranked table; CSV; receipt
Edits	Up to 5,000 reference and edited pairs (CSV or JSON), with optional family; Siraj or Kreimer 0–72 h toggle	Scores, top contributing JASPAR motifs (Kreimer), family-diverse panel of requested size; CSV; receipt
Verify	A receipt file	Three checks: digest, identity of model and endpoint, replay of every stored result

ing unequal-length insertions and deletions) compile into `brainc` v2 Development Modules, and Studio exposes per-edit JASPAR motif attributions alongside the seven-timepoint curve. Because GSE188264 estimates were preprocessed with MPRAnalyze prior to splitting and the 118 test families had been examined during earlier k-mer studies, this result is retrospective and conditional (`biological_development_accepted = False`).

5.7 Intended use

Both predictors are meant for prioritizing edits within their studied assay or exploring sequence hypotheses—for example, contrasting passive MPRA predictions against coupled ODE fate flips in Section 4.4. Neither predictor is validated for cross-assay or endogenous clinical effect sizes without fresh confirmation.

6 Epigenesis Studio

Studio (`studio/`, command `epigenesis-studio` or `epigenesis studio`) is a local desktop application in Version 0.2.0. Its server listens on 127.0.0.1 only and rejects requests whose Host header is not local. The interface is in English and Bulgarian, with a switch in the sidebar. It has five tabs (Table 8).

Simulate and Enhancer Circuit run the repository backends of Section 4 live; there is no second simulator. Both require the repository checkout, and without it they are disabled while the remaining tabs still work.

Scan and Edits run either the Siraj Huber-linear checkpoint or the Kreimer 0–72 h motif-temporal checkpoint of Section 5. Scan scores every single-base change in a window around the position, which equals whole-sequence scoring (tested to 10^{-12}) because linear scores depend only on the local k-mer and motif radius (18 bases either side for Siraj and Kreimer). Edits scores full reference/edited pairs (including indels and MNVs) and selects a family-diverse panel.

A receipt (`epigenesis-studio-receipt-v1`) holds the model name, its identity hash, the endpoint, the normalized request, the result and a SHA-256 digest over the RFC 8785 canonical JSON of all of it. Verify checks the digest, that the model name, identity and endpoint match the installation, and that

rerunning the stored request reproduces every stored result within 10^{-9} (Table 6).

Simulate and Enhancer Circuit compile their alleles into sealed `brainc` Development Modules; Scan and Edits seal canonical JSON receipts directly. All 32 Studio tests pass, covering local-window versus whole-sequence parity (Siraj and Kreimer), panel selection, cryptic bifurcations and second-site rescues in Enhancer Circuit, all three receipt checks, input limits and refusals, HTTP host/path guards, bilingual EN/BG key parity, and CLI parity.

7 Limits and open questions

Simulation. All mechanistic results depend on the declared assumptions of Section 4.3. In-vitro binding ratios or Boltzmann partition sums enter as relative affinity multipliers without measured intracellular calibration, and embryo counts (Peterson et al. 2012) are displayed rather than scored. While Version 1.7 generalizes enhancer length, transcription-factor PWM choice (Hallikas GLI2/GLI3, 532 JASPAR 2024 motifs, or custom matrices), and arbitrary N-gene `CircuitSpec` wiring, nucleosome positioning, co-factor complexes, 3D chromatin looping, and multicellular organism phenotypes remain outside the model.

Predictor. Siraj internal gains are preliminary because development reused TUNING, the ten-edit bound remains open, and in thirteen frozen cross-study transfer tests (Appendix B) no model earned a transfer label (nearest-TSS 0.460 beat all sequence models on credible sets). Kreimer 0–72 h motif-temporal results improve retrospectively over k5 and dinucleotide baselines on GSE188264 yet remain conditional on published MPRAnalyze preprocessing and previously examined test families. Cross-context variant prediction therefore remains open.

Evaluation. The hashes of the frozen procedure are local; they guard against accidents rather than deliberate rewriting, and the thirteen transfer tests are uncorrected for multiplicity. Several external sets were exposed to earlier models or share laboratory lineages. The predictor endpoint is reporter activity or expression rather than cellular fate or morphology.

Next steps. (a) Compare simulated cryptic bifurcations and second-site epistatic rescues against measured endogenous CRISPR edits, replacing assumed affinity parameters with biochemical measurements. (b) Score the sealed Siraj RESERVE partition once for one prespecified candidate. (c) Collect fresh, unexposed endogenous-locus and multi-context datasets on calibrated scales.

8 Conclusion

Epigenesis Version 1.7 links DNA sequences under explicit, independently replayed interpretations to both mechanistic N-gene fate simulations and interpretable sequence predictors. In a single 9-base GLI site, a substitution flips Nkx2.2 from 0.940 to 0.0085 while reversion restores the reference exactly; across full 171-base enhancers and multi-TF circuits, Boltzmann partition scanning uncovers cryptic bifurcations where passive reporters barely move yet coupled feedback switches cell identity, alongside candidate epistatic rescues. Together with the Siraj and Kreimer 0–72 h predictors and bilingual Studio receipts, every run remains computationally auditable while keeping its biological scope and cross-context boundaries explicit.

9 Availability and statements

Code and data. `https://github.com/dttdrv/epigenesis`. Code is Apache-2.0. Bundled data and third-party components keep their licenses and attributions: JASPAR 2024 is CC BY 4.0, the 80K-Analysis notice is in `examples/neural-predictor/LICENSE.Salomon`, and the bundled attention predictor uses Interlace (see `NOTICE.md`).

Running it. Python 3.11 or later runs the compiler, unified CLI (`python3.11 epigenesis_cli.py check --full`), and core experiments using the standard library. Studio (`uv tool install --editable studio`

`--with-editable .`) and the sequence predictors require NumPy; CX3D construction bridging uses Java 21 and `patch`.

Figure data. `docs/paper/figure-patterning.csv` records the 9-bp GAACACCCA GLI2 simulation ($S = 0.2$, 20 model units). `docs/paper/figure-data.py` rebuilds `figure-scan.csv` and `figure-studies.csv` and prints the Studio demonstration numbers of Section 5. Table 3 numbers reproduce directly via `examples/neural-circuit/circuit.py` (`demo`, `benchmark`, and `scan --circuit tri-lineage-switch`); Section 5.6 numbers reproduce via `python3.11 tests/verify.py neural-regulation-motif`.

Frozen tests. Plans, hashes, scripts and results are in `research/frozen-tests/` (Appendix C). `python3.11 research/check_freeze.py` recomputes the hashes. The Siraj RESERVE partition remains sealed. The Salomon and Kosicki confirmations are spent.

Data without a journal article. Gene Expression Omnibus series GSE244011, GSE211045, GSE225815, GSE221384, GSE152404 and GSE188264 (Kreimer et al. 2022); 80K-Analysis (`https://github.com/kircherlab/80K-Analysis`); Interlace (`https://github.com/MisulOrg/Interlace`); CX3D 0.03 (`https://tilde.ini.uzh.ch/~amw/seco/cx3d/cx3d-0.03.zip`); GENCODE v39 (Frankish et al. 2021); UCSC hg38 and phyloP100way tracks.

Responsibility. The author is responsible for the content. Every number comes from the executable checks, result files listed in Appendix C, example folders and the repository README. Competing interests and funding: to be completed by the author before submission.

References

Agarwal V, Inoue F, Schubach M, et al. (2025). Massively parallel characterization of transcriptional regulatory elements. *Nature* 639:411–420. doi:10.1038/s41586-024-08430-9

Ashuach T, Fischer DS, Kreimer A, et al. (2019). MPRAnalyze: statistical framework for massively parallel reporter assays. *Genome Biology* 20:183. doi:10.1186/s13059-019-1787-z

Avsec Ž, Agarwal V, Visentin D, et al. (2021). Effective gene expression prediction from sequence by integrating long-range interactions. *Nature Methods* 18:1196–1203. doi:10.1038/s41592-021-01252-x

Aygün N, Elwell AL, Liang D, et al. (2021). Brain-trait-associated variants impact cell-type-specific gene regulation during neurogenesis. *American Journal of Human Genetics* 108:1647–1668. doi:10.1016/j.ajhg.2021.07.011

Balaskas N, Ribeiro A, Panovska J, et al. (2012). Gene regulatory logic for reading the Sonic Hedgehog signaling gradient in the vertebrate neural tube. *Cell* 148:273–284. doi:10.1016/j.cell.2011.10.047

Beal J, Lu T, Weiss R. (2011). Automatic compilation from high-level biologically-oriented programming language to genetic regulatory networks. *PLoS ONE* 6:e22490. doi:10.1371/journal.pone.0022490

Bergmann FT, Adams R, Moodie S, et al. (2014). COMBINE archive and OMEX format: one file to share all information to reproduce a modeling project. *BMC Bioinformatics* 15:369. doi:10.1186/s12859-014-0369-z

Bilitchenko L, Liu A, Cheung S, et al. (2011). Eugene – a domain specific language for specifying and constraining synthetic biological parts, devices, and systems. *PLoS ONE* 6:e18882. doi:10.1371/journal.pone.0018882

Cai Y, Lux MW, Adam L, Peccoud J. (2009). Modeling structure-function relationships in synthetic DNA sequences using attribute grammars. *PLoS Computational Biology* 5:e1000529. doi:10.1371/journal.pcbi.1000529

Chen KM, Wong AK, Troyanskaya OG, Zhou J. (2022). A sequence-based global map of regulatory activity for deciphering human genetics. *Nature Genetics* 54:940–949. doi:10.1038/s41588-022-01102-2

Cohen M, Page KM, Perez-Carrasco R, Barnes CP, Briscoe J. (2014). A theoretical framework for the regulation of Shh morphogen-controlled gene expression. *Development* 141:3868–3878. doi:10.1242/dev.112573

Deng C, Whalen S, Steyert M, et al. (2024). Massively parallel characterization of regulatory elements in the developing human cortex. *Science* 384:eadh0559. doi:10.1126/science.adh0559

Frankish A, et al. (2021). GENCODE 2021. *Nucleic Acids Research* 49:D916–D923. doi:10.1093/nar/gkaa1087

Fromer M, Roussos P, Sieberts SK, et al. (2016). Gene expression elucidates functional impact of polygenic risk for schizophrenia. *Nature Neuroscience* 19:1442–1453. doi:10.1038/nn.4399

Gneiting T, Raftery AE. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association* 102:359–378. doi:10.1198/016214506000001437

Gong et al. (2020). The requirement of Sox2 for the spinal cord motor neuron development of zebrafish. *Frontiers in Molecular Neuroscience* 13:34. doi:10.3389/fnmol.2020.00034

GTEx Consortium (2020). The GTEx Consortium atlas of genetic regulatory effects across human tissues. *Science* 369:1318–1330. doi:10.1126/science.aaz1776

Guo et al. (2023). Integrative analyses highlight functional regulatory variants associated with neuropsychiatric diseases. *Nature Genetics* 55:1876–1891. doi:10.1038/s41588-023-01533-5

Hallikas O, Palin K, Sinjushina N, et al. (2006). Genome-wide prediction of mammalian enhancers based on analysis of transcription-factor binding affinity. *Cell* 124:47–59. doi:10.1016/j.cell.2005.10.042

Herrera-Delgado E, Briscoe J, Sollich P. (2020). Tractable nonlinear memory functions as a tool to capture and explain dynamical behaviors. *Physical Review Research* 2:043069. doi:10.1103/PhysRevResearch.2.043069

Huber PJ. (1964). Robust estimation of a location parameter. *Annals of Mathematical Statistics* 35:73–101. doi:10.1214/aoms/1177703732

Jaffe AE, Straub RE, Shin JH, et al. (2018). Developmental and genetic regulation of the human cortex transcriptome illuminate schizophrenia pathogenesis. *Nature Neuroscience* 21:1117–1125. doi:10.1038/s41593-018-0197-y

Jerber J, Seaton DD, Cuomo ASE, et al. (2021). Population-scale single-cell RNA-seq profiling across dopaminergic neuron differentiation. *Nature Genetics* 53:304–312. doi:10.1038/s41588-021-00801-6

Kang HJ, Kawasawa YI, Cheng F, et al. (2011). Spatio-temporal transcriptome of the human brain. *Nature* 478:483–489. doi:10.1038/nature10523

Kelley DR, Snoek J, Rinn JL. (2016). Basset: learning the regulatory code of the accessible genome with deep convolutional neural networks. *Genome Research* 26:990–999. doi:10.1101/gr.200535.115

Kelley DR, Reshef YA, Bileschi M, et al. (2018). Sequential regulatory activity prediction across chromosomes with convolutional neural networks. *Genome Research* 28:739–750. doi:10.1101/gr.227819.117

Kerimov N, Hayhurst JD, Peikova K, et al. (2021). A compendium of uniformly processed human gene expression and splicing quantitative trait loci. *Nature Genetics* 53:1290–1299. doi:10.1038/s41588-021-00924-w

Kircher M, Xiong C, Martin B, et al. (2019). Saturation mutagenesis of twenty disease-associated regulatory elements at single base-pair resolution. *Nature Communications* 10:3583. doi:10.1038/s41467-019-11526-w

Kreimer A, Ashuach T, Inoue F, et al. (2022). Massively parallel reporter perturbation assays uncover temporal regulatory architecture during neural differentiation. *Nature Communications* 13:1504. doi:10.1038/s41467-022-28659-0

Lee S, McAfee JC, Lee J, et al. (2025). Massively parallel reporter assay investigates shared genetic variants of eight psychiatric disorders. *Cell* 188:1409–1424.e21. doi:10.1016/j.cell.2024.12.022

Majoros WH, Kim YS, Barrera A, et al. (2020). Bayesian estimation of genetic regulatory effects in high-throughput reporter assays. *Bioinformatics* 36:331–338. doi:10.1093/bioinformatics/btz545

Matoba N, McAfee JC, Krupa O, et al. (2026). Massively parallel assessment of gene regulatory activity at human cortical-structure-associated variants. *Nature Neuroscience*. doi:10.1038/s41593-026-02454-2

McAfee JC, Lee S, Lee J, et al. (2023). Systematic investigation of allelic regulatory activity of schizophrenia-associated common variants. *Cell Genomics* 3:100404. doi:10.1016/j.xgen.2023.100404

Melnikov A, Murugan A, Zhang X, et al. (2012). Systematic dissection and optimization of inducible enhancers in human cells using a massively parallel reporter assay. *Nature Biotechnology* 30:271–277. doi:10.1038/nbt.2137

Miller JA, Ding SL, Sunkin SM, et al. (2014). Transcriptional landscape of the prenatal human brain. *Nature* 508:199–206. doi:10.1038/nature13185

Myint L, Avramopoulos DG, Goff LA, Hansen KD. (2019). Linear models enable powerful differential activity analysis in massively parallel reporter assays. *BMC Genomics* 20:209. doi:10.1186/s12864-019-5556-x

Ng B, White CC, Klein HU, et al. (2017). An xQTL map integrates the genetic architecture of the human brain's transcriptome and epigenome. *Nature Neuroscience* 20:1418–1426. doi:10.1038/nn.4632

Liu et al. (2025). Predicting the DNA binding specificity of transcription factor mutants using family-level biophysically interpretable machine learning. *Nucleic Acids Research* 53:gkaf831. doi:10.1093/nar/gkaf831

Nielsen AAK, Der BS, Shin J, et al. (2016). Genetic circuit design automation. *Science* 352:aac7341. doi:10.1126/science.aac7341

NIST (2015). *Secure Hash Standard (SHS)*, FIPS 180-4. `https://csrc.nist.gov/pubs/fips/180-4/upd1/final`

Peterson KA, Nishi Y, Ma W, et al. (2012). Neural-specific Sox2 input and differential Gli-binding affinity provide context and positional information in Shh-directed neural patterning. *Genes & Development* 26:2802–2816. doi:10.1101/gad.207142.112

Pollard KS, Hubisz MJ, Rosenbloom KR, Siepel A. (2010). Detection of nonneutral substitution rates on mammalian phylogenies. *Genome Research* 20:110–121. doi:10.1101/gr.097857.109

Ramasamy A, Trabzuni D, Guelfi S, et al. (2014). Genetic variability in the regulation of gene expression in ten

regions of the human brain. *Nature Neuroscience* 17:1418–1428. doi:10.1038/nn.3801

Rauluseviciute I, Riudavets-Puig R, Blanc-Mathieu R, et al. (2024). JASPAR 2024: 20th anniversary of the open-access database of transcription factor binding profiles. *Nucleic Acids Research* 52:D174–D182. doi:10.1093/nar/gkad1059

Roadmap Epigenomics Consortium, Kundaje A, Meuleman W, et al. (2015). Integrative analysis of 111 reference human epigenomes. *Nature* 518:317–330. doi:10.1038/nature14248

Rummel CK, Gagliardi M, Ahmad R, et al. (2023). Massively parallel functional dissection of schizophrenia-associated noncoding genetic variants. *Cell* 186:5165–5182.e33. doi:10.1016/j.cell.2023.09.015

Rundgren A, Jordan B, Erdtman S. (2020). *JSON Canonicalization Scheme (JCS)*, RFC 8785. `https://www.rfc-editor.org/rfc/rfc8785`

Salomon K, Deng C, Dash PM, et al. (2026). Massively parallel characterization and predictive modelling of neuronal regulatory variation. *bioRxiv* (preprint, posted 17 July 2026). doi:10.64898/2026.07.16.738760

Schwartzentruber J, Foskolou S, Kilpinen H, et al. (2018). Molecular and functional variation in iPSC-derived sensory neurons. *Nature Genetics* 50:54–61. doi:10.1038/s41588-017-0005-8

Seitz EE, McCandlish DM, Kinney JB, Koo PK. (2024). Interpreting cis-regulatory mechanisms from genomic deep neural networks using surrogate models. *Nature Machine Intelligence* 6:701–713. doi:10.1038/s42256-024-00851-5

Siraj L, et al. (2026). Functional dissection of complex trait variants at single-nucleotide resolution. *Nature*. doi:10.1038/s41586-026-10121-6

Tareen A, Kooshkbaghi M, Posfai A, et al. (2022). MAVE-NN: learning genotype-phenotype maps from multiplex assays of variant effect. *Genome Biology* 23:98. doi:10.1186/s13059-022-02661-7

Tewhey R, Kotliar D, Park DS, et al. (2016). Direct identification of hundreds of expression-modulating variants using a multiplexed reporter assay. *Cell* 165:1519–1529. doi:10.1016/j.cell.2016.04.027

Traets JJ, van der Burght SN, Rademakers S, et al. (2021). Mechanism of life-long maintenance of neuron identity despite molecular fluctuations. *eLife* 10:e66955. doi:10.7554/eLife.66955

Vaishnav ED, de Boer CG, Molinet J, et al. (2022). The evolution, evolvability and engineering of gene regulatory DNA. *Nature* 603:455–463. doi:10.1038/s41586-022-04506-6

Walker RL, Ramaswami G, Hartl C, et al. (2019). Genetic control of expression and splicing in developing human brain informs disease mechanisms. *Cell* 179:750–771.e22. doi:10.1016/j.cell.2019.09.021

Wang G, Sarkar A, Carbonetto P, Stephens M. (2020). A simple new approach to variable selection in regression, with application to genetic fine mapping. *Journal of the Royal Statistical Society Series B* 82:1273–1300. doi:10.1111/rssb.12388

Weiss CV, Harshman L, Inoue F, et al. (2021). The cis-regulatory effects of modern human-specific variants. *eLife* 10:e63713. doi:10.7554/eLife.63713

Wilson ML, et al. (2016). Genotype Specification Language. *ACS Synthetic Biology* 5:471–478. doi:10.1021/acssynbio.5b00194

Young AMH, Kumasaka N, Calvert F, et al. (2021). A map of transcriptional heterogeneity and regulatory variation in human microglia. *Nature Genetics* 53:861–868. doi:10.1038/s41588-021-00875-2

Zhou J, Troyanskaya OG. (2015). Predicting effects of noncoding variants with deep learning–based sequence model. *Nature Methods* 12:931–934. doi:10.1038/nmeth.3547

Zubler F, Douglas R. (2009). A framework for modeling the growth and development of neurons and networks. *Frontiers in Computational Neuroscience* 3:25. doi:10.3389/neuro.10.025.2009

Zubler F, Hauri A, Pfister S, et al. (2011). An instruction language for self-construction in the context of neural networks. *Frontiers in Computational Neuroscience* 5:57. doi:10.3389/fncom.2011.00057

Zubler F, Hauri A, Pfister S, et al. (2013). Simulating cortical development as a self constructing process: a novel multi-scale approach combining molecular and physical aspects. *PLoS Computational Biology* 9:e1003173. doi:10.1371/journal.pcbi.1003173

A Frozen evaluation procedure

Every test in Appendix B follows one procedure.

1. **Plan.** A plan file fixes the question, populations, models, baselines, metrics, bootstrap, decision gates and the label that a pass would earn.
2. **Hash.** `FREEZE.json` records SHA-256 digests of the plan, scripts and inputs before any outcome is

read. A script that never sees held-out outcomes makes the predictions, and their digest is added.

3. **Qualify.** The scorer runs on a synthetic positive, where the label must come out true, and on a synthetic null, where it must not. A scorer that fails either run is repaired first.
4. **Score once.** The scorer verifies the digests, scores the sealed predictions and writes one result. A changed model needs a new plan and, if the data are spent, new data.
5. **Judge.** The label is earned only if every gate passes. For example, the lower bound of the candidate exceeds the null, the candidate beats the best fixed baseline and it beats a model trained on other data.

The resampling unit is the sequence family, gene or credible set. Units are resampled as whole copies in 2,000 bootstrap draws. Intervals are 5th to 95th percentiles, so the lower end is a one-sided 95% bound. Correlations are family-weighted Pearson r, with studies weighted equally. Selection quality is signed utility. It is the observed effect, signed by the predicted direction, of the edits that a model selects (one per family) for a panel of 10, 50 or 200. Random selection scores near zero. Direction is accuracy, with ties counted as one half.

Data fall into four classes. *Sealed* data are unopened (the Siraj RESERVE partition). *Spent* data were scored once and cannot confirm later models (the Salomon and Kosicki confirmations). *Exposed* data guided development (the Siraj TUNING partition, earlier reporter studies, and each eQTL test set after its first scoring). Results on exposed data carry an adaptivity caveat. *Fresh* data are new to every model in the test.

B Research record: sequence-based transfer tests

The tests ask whether sequence alone predicts what an unseen DNA change does in neural cells. They are the evidence behind the transfer boundary of Section 5.5.

B.1 Task, data and models

Task. Given a reference and an edited sequence, predict the alternate-minus-reference reporter effect (a $\log_2$ ratio). For natural variants, predict the direction of the effect on expression.

Training data. The main set is the SK-N-SH reporter screen of Siraj et al. (2026) (Section 5.1). WTC11-NGN2 neuron data (Salomon et al. 2026) supply activity and effects for one model class. Within each study, outcomes are centered and scaled to unit family-weighted standard deviation before pooling.

Evaluation data. Six neural reporter studies with allelic effects: Lee et al. (GSE244011; 5,916 variants; Lee et al. 2025), McAfee et al. (GSE211045; 1,682; McAfee et al. 2023), Matoba et al. (4,013; Matoba et al. 2026), Rummel et al. (GSE225815; 611; Rummel et al. 2023), Guo et al. (GSE221384; 748; Guo et al. 2023) and Weiss et al. (GSE152404; 9,247; Weiss et al. 2021). The Matoba set is the project's TRAIN partition. Section 5.1 describes the held-out sets, and Table 7 gives their sizes. A family is a connected component of variants within 2,000 bases (a locus for Matoba).

Models. *k5* is ridge regression on the difference in canonical 5-mer counts (512 coefficients). *S1* is k5 trained on the Siraj screen. *S2* adds the difference in JASPAR 2024 motif potential (Rauluseviciute et al. 2024). It is the standing reference. *Huber-linear* fits the same 512 word and 532 motif coefficients with a Huber loss and is the delivered model. *S3* and *H3* add a quadratic motif-context term. They are trained on Siraj and on the other studies, respectively. *H2* is k5 plus motifs, trained on the other studies. *Cfit* is a fitted blend of study-trained and Siraj-trained models. Further models are ridge regression on 611 Sei scores (Chen et al. 2022) and a convolutional network trained on pooled effects. There is also an activity-transfer model learned from WTC11-NGN2 measurements (motivated by Seitz et al. 2024; Tareen et al. 2022). The last is a zero-shot model that multiplies the change in motif occupancy by transcription-factor expression in 8 to 13 post-conception-week neocortex (Kang et al. 2011; Miller et al. 2014).

B.2 Earlier negative experiments

Two experiments outside the Siraj series ended in negative results. One is a frozen k-mer predictor for H1 neural induction (`examples/neural-regulation/regulation.py`). Its frozen model lowered mean squared error by 20.97% against zero effect but failed the required comparison with a dinucleotide model (Section 5.6 presents the Version 1.7 motif-temporal follow-up that surpasses both k5 and dinucleotide baselines retrospectively). The other is a benchmark of the FamilyCode homeodomain model (Liu et al. 2025) on *Nkx2-2*, which failed.

B.3 Measurement checks

Forecasting omitted repeats. In a neuronal-assay pilot we omitted one RNA repeat at a time and forecast its allelic fraction from the others, with the BIRD model (Majoros et al. 2020) or by pooling with an analytic Beta-binomial, and scored the forecasts with the continuous ranked probability score (CRPS; Gneiting and Raftery 2007). Over twelve omissions in four TRAIN components, the mean CRPS was 0.0549 for BIRD and 0.0622 for pooling, 11.7% lower. The conditional 99% simulation interval for the gain is 0.0044 to 0.0098 and covers Monte Carlo error only, for this selected cohort. All four component means improve, and the one omission that favors pooling is kept. BIRD and pooling also differ in priors, so the comparison does not isolate repeat variation as the cause. A DNA-normalized follow-up lowered mean CRPS by 15.8% against pooling, but a post hoc zero-effect forecast scored better.

Forecasting later responses. In the neural-induction assay of Kreimer et al. (2022), ridge regression on four early measurements lowered error by 24% against carrying the 12-hour value forward, in 351 held-out families, and matched a simpler averaging control. Two zero-shot pretrained forecasting models did no better. A raw-count test that holds out families and infection cultures gave a mean squared error of 0.3213 against 0.3221 for zero change (0.26% better). All four gain intervals include zero.

Counting noise. In GSE225815, 83% of variants have one allele with fewer than ten RNA reads, so the log ratio is set mainly by the pseudocount and the DNA-count imbalance. The high agreement between replicate halves (0.98) comes from the shared DNA count and says nothing about reproducible biology: on the 124 well-measured variants, the Siraj-trained head scores $r = -0.09$. The other three count-based studies have negligible counting noise (median Poisson variance below 0.002 $\log_2^2$). We therefore dropped GSE225815 from the primary group in later tests and kept it as a sensitivity analysis.

B.4 Reporter effects

Within one assay. Section 5.3 gives the internal comparison of Huber-linear with k5. Two earlier confirmations on spent data, for earlier fitted predictors of this series, failed. The Kosicki confirmation (3,840 edits in 50 families) gave an error of 0.016958 for the fixed predictor against 0.016919 for zero effect. Its ten-edit utility was −0.038, against +0.023 for random selection. The Salomon confirmation (7,372 edits in 2,938 families) gave an error of 0.028957 against 0.028998 for k5, and a ten-edit utility of 0.170 against 0.153. All seven point comparisons favored the model, and every interval included zero.

Across studies. Table 9 lists the tests that apply a fixed model to other studies. The reference S2 is trained on Siraj alone. It has $r \approx 0.047$ (S1 has 0.047 over all six studies, interval 0.026 to 0.069), that is $r^2 \approx 0.002$. It is positive in four of the five studies other than GSE225815. After the count check, these five studies are the primary group. The six-study group is a sensitivity analysis.

Test 8 is the delivered model (Section 5.5, which also states the comparison with predicting zero effect). Per-study correlations of S2 range from −0.014 (Guo et al.) to 0.115 (Weiss et al.). In an exploratory, outcome-selected analysis, the sign of S2 is right for 53% to 57% of the top 10% largest effects in the Lee, Matoba and Weiss studies, against 50% overall. This is a lead only. Evolutionary conservation (phyloP100way; Pollard et al. 2010) does not help. Its correlation with absolute effect averages 0.017, and multiplying S2 by conservation lowers the mean r from 0.047 to 0.034.

Table 9. Cross-study tests. r is family-weighted Pearson correlation with studies weighted equally (5th to 95th percentiles). Each test was scored once, 19 to 20 October 2026. No test earned its label.

#	Candidate and population	Candidate r	Reference r
1	H2, trained on the two other studies; McAfee and Matoba folds	0.015 (−0.024, 0.054)	S1 0.060
2	Cfit blend; three newly scored studies	0.046 (0.009, 0.082)	S2 0.050
3	Sei ridge; six studies (blend with S2: 0.041)	−0.003 (−0.024, 0.019)	S2 0.047
4	Quadratic context head S3; six studies	0.046	S2 0.047
5	WTC11-NGN2 activity transfer; six studies	−0.010 (−0.029, 0.011)	S2 0.047
6	Pooled convolutional network; five studies	0.014 (−0.008, 0.035)	S2 0.047
7	Zero-shot expressed-TF binding; five studies	0.0015	S2 0.0465
8	Huber-linear, fixed; five studies	0.0490	S2 0.0465

Table 10. Direction accuracy on held-out fine-mapped eQTLs; resampling by gene, 5th to 95th percentiles.

Test set	n	M4	M5	M5 against best baseline
Fetal neocortex (Walker et al. 2019)	792	0.547	0.566	B3: +0.028 (−0.009, +0.065)
Progenitors (Aygün et al. 2021)	229	0.528	0.646	B2: +0.066 (0.013, 0.117)
Neurons (Aygün et al. 2021)	80	0.575	0.600	intervals wide
Differentiating neurons (Jerber et al. 2021)	320	0.534	0.603	B2: +0.003 (−0.037, +0.045)
Sensory neurons (Schwartzentruber et al. 2018)	519	0.538	0.518	B2 0.499 (PIP ≥ 0.1)
Microglia (Young et al. 2021)	272	n/a	0.570	reporter head 0.574

B.5 Natural variants

Direction of expression effects. The training data were adult-brain fine-mapped QTLs: GTEx brain regions, BrainSeq, CommonMind, ROSMAP and Braineac2 in the eQTL Catalogue (GTEx Consortium 2020; Jaffe et al. 2018; Fromer et al. 2016; Ng et al. 2017; Ramasamy et al. 2014; Kerimov et al. 2021). They comprise 14,617 variant-gene pairs with a posterior inclusion probability (PIP) of at least 0.5, fine-mapped with SuSiE (Wang et al. 2020). Folds hold out whole chromosomes. Tests used fetal and neural-derived datasets. M4 uses Sei chromatin scores by distance bin to the transcription start site (TSS). M5 adds GC change, the local k5 and the reporter head. Baselines are B1 (the alternate allele gains GC), B2 (k5 logistic regression) and B3 (the Siraj-trained reporter head, unfitted).

Pooled over the four neural-derived sets at PIP ≥ 0.5 (626 pairs), M5 reaches 0.599 (0.566, 0.630) against 0.567 for B2. The largest single-set gain, in the progenitors, did not repeat in the Jerber or sensory-neuron sets. No model is consistently above the k-mer baseline. The three labels (fetal-brain direction, confirmation in differentiating neurons, replication in neural-derived cells) were not earned.

Which variant in a credible set is causal? A conditional-logit ranking of credible-set members was trained on 5,836 adult-brain sets (27,875 members) with leave-chromosome-out folds. We scored top-1 accuracy on 439 fetal-neocortex sets (Walker et al. 2019) and 420 neural-derived sets (Table 11).

No model beats the nearest-TSS rule in fetal neocortex. In the neural-derived pool, M4 exceeds distance bins by +0.045 (+0.005, +0.085). It does not clearly exceed B1 (M5 − B1 = +0.019, interval −0.029 to +0.067). Conservation carries no signal: phyloP picks the member with the highest fine-mapping probability no better than chance (pairwise AUC 0.495 in training sets, 0.503 in the Walker sets). Roadmap DNase accessibility (Roadmap Epigenomics Consortium 2015) does not help in fetal neocortex. The two accessibility models reach 0.417 and 0.420, below M4 (0.435) and nearest-TSS (0.460). Neither label was earned.

Table 11. Top-1 accuracy of credible-set ranking (5th to 95th percentiles).

Method	Fetal neocortex	Neural-derived pool
Random member	0.355	0.328
Nearest TSS (B1)	**0.460** (0.420, 0.499)	0.402 (0.362, 0.442)
Distance bins (B2)	0.433 (0.403, 0.463)	0.381 (0.351, 0.411)
Highest phyloP (B3)	0.355 (0.320, 0.392)	0.333 (0.294, 0.373)
Distance + Sei (M4)	0.435 (0.396, 0.474)	**0.426** (0.386, 0.464)
Distance + Sei + phyloP (M5)	0.419 (0.381, 0.458)	0.421 (0.381, 0.461)

B.6 Summary

Across the thirteen tests, the Siraj-trained head carries a small, stable cross-study signal (r near 0.05) that nothing tried adds to. Signals that look better in one set do not repeat in the next. Simple baselines (nearest TSS, k-mer counts, the reporter head) are hard to beat on natural variants.

C Frozen tests and where they live

Each directory under `research/frozen-tests/` holds the plan (`PLAN.md`), digests (`FREEZE.json`), `scripts/`, small result files and `RESULTS.md`. The original tree is `build/neural-predictor-reviewed-20261005/`, which is not versioned.

#	Test	Directory
1	Leave-one-study-out, three studies	`hnp-loso-20261019`
2	Combined predictor, six studies	`hnp-loso6-20261019`
3	Sei ridge	`sei-features-20261019`
4	Quadratic context head	`context-loso6-20261019`
5	Activity-transfer model	`activity-transfer-loso6-20261019`
6	Pooled convolutional network	`cnn-loso6-20261019`
7	Zero-shot expressed-TF binding	`mechanism-zeroshot-20261019`
8	Huber-linear transfer	`huber-transfer-20261020`
9	eQTL direction, fetal neocortex	`eqtl-brain-20261019`
10	eQTL direction, Jerber	`eqtl-brain-20261019` (`PLAN_CONFIRM.md`)
11	eQTL direction, sensory neurons and microglia	`eqtl-brain-20261019` (`PLAN_REP.md`)
12	Credible-set ranking	`eqtl-causal-20261019` (`PLAN.md`)
13	Credible-set ranking with accessibility	`eqtl-causal-20261019` (`PLAN2.md`)

An exploratory, unfrozen analysis of conservation and direction is in `constraint-factor-20261019`.